1 Propose candidates

SAM 3, frozen



image

"Locate the handle of the mug."

phrases $\phi(x)$

- the handle of the mug
- the mug

SAM 3 · point grid
32×32, query-free, ≤ 256

SAM 3 · text
top 3 per phrase

Candidate pool

N = 260



grid masks SAM 3 text instances (#rank)

Containment tree



$\text{parent}(v) = \arg\min_u |m_u| \;\text{ s.t. }\; |m_u \cap m_v| \geq 0.8\,|m_v|$

e.g. mug ⊃ handle; used between mask tokens

2 Mask tokenization

candidate m_v



Appearance $a_v = \sum_j \tilde{c}_{v,j}\, e_j$
what it shows: the MLLM's image tokens under the mask

Geometry $g_v \in \mathbb{R}^9$
where it is: box, area, depth in the tree

Source $q_v \in \mathbb{R}^5$
where it came from: grid or SAM 3 text (rank, score)

concatenate

MLP + RMS norm

$\hat{x}_v = \text{MLP}\big([\,a_v;\, g_v;\, q_v\,]\big)$

position (mRoPE): mask centroid



like a word token: content + position + type

3 The MLLM selects

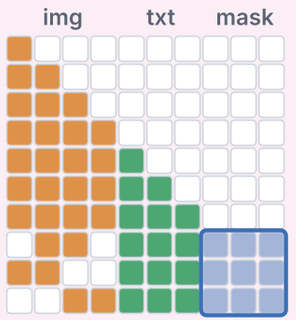
Qwen3-VL-4B + LoRA



image tokens query mask tokens (one pass)

Large language model
Qwen3-VL-4B + LoRA, 64.2M trainable parameters

Region-restricted attention (every layer)



- a mask token sees only its own image region, plus the query
- image and query tokens never see the mask tokens
- between mask tokens: a learned bias $b_{\ell,\,\text{rel}(u,v)}$ by tree relation

Selection head

$v^\star = \arg\max_v\; s_v$

$s_v = \lambda\, \cos\big(\psi_q(h_{\langle\text{SEL}\rangle}),\, \psi_k(z_v)\big) + \beta$

scores over the pool



answer $m_{v^\star}$

